

SUMS Proofs Workshop: Winter 2026 Solutions

Note: Problems marked with (*) are difficult. (**) indicates *very difficult*.

Problems from Fall's Workshop

PROBLEM 1 Show that $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for every $n \in \mathbb{Z}^+$.

PROBLEM 2 Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$.

PROBLEM 3 (*) Assume that the following statement is true: if Micah dislikes pineapple pizza then he likes pineapple pizza. Does he like pineapple pizza?

PROBLEM 4 (**) The *Fibonacci sequence* is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2}$$

for $n \geq 2$. Show that every positive integer can be written as the sum of distinct Fibonacci numbers. Distinct means the sum cannot have repeated values; For example, $2 = 1 + 1$ is not a valid sum even though $F_1 = 1$ and $F_2 = 1$. But $2 = 2 = F_3$ is.

PROBLEM 5 (**) What is the minimum number of breaks to break up a chocolate bar made of $m \times n$ little squares into those mn little squares?

Starter Problems

PROBLEM 6 Consider the following statement: For all positive integers x , $x^2 + x + 41$ is prime. Is the statement true or false? Prove it.

PROBLEM 7 Prove that no integer n , when squared, ends in 2, 3, 7, or 8.

PROBLEM 8 Let $n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$. Prove that $n!$ is divisible by $2^{n/2}$ for all even positive integers n .

PROBLEM 9 Prove that if a divides b and b divides a , then $|a| = |b|$. Is it true that $a = b$ too? Why or why not?

PROBLEM 10 Prove that if $A \subseteq B$ then $B^c \subseteq A^c$, where A^c is the complement of A .

PROBLEM 11 Show that in any group of n people, where each person is friends with some of the other $n - 1$, there are always two people with the same number of friends.

PROBLEM 12 You have a pile of 100 coins. Every move, you may replace a pile of size n with two piles of a and b coins, where $a + b = n$ and $a \neq b$. Can you end with 100 piles of 1 coin? Prove or disprove.

PROBLEM 13 Let $\gcd(a, b)$ denote the greatest common divisor between integers a and b . Prove that $\gcd(a, b) = \gcd(a, b - a)$ for all integers a and b .

PROBLEM 14 Show that $2 + 4 + 6 + \cdots + 2n = n(n + 1)$. Can you do it without induction?

PROBLEM 15 Prove that if $A \times B = A \times C$ and $A \neq \emptyset$, then $B = C$.

Harder Problems

PROBLEM 16 (*) Prove that if $2^n - 1$ is prime, then n is prime. Hint:

$$2^{ab} - 1 = (2^a - 1)(2^{a(b-1)} + 2^{a(b-2)} + \cdots + 1).$$

PROBLEM 17 (*) Prove that if $a \equiv b \pmod{m}$ then $a^k \equiv b^k \pmod{m}$ for all positive integer k .

PROBLEM 18 (*) Prove $\sqrt{6}$ is irrational.

PROBLEM 19 (*) Show that every squared integer can be written as the sum of consecutive odd numbers.

PROBLEM 20 (*) Start with the number 0. Every move, you can add 3 or 5 to it. Show that, given enough moves, you can make every integer ≥ 8 , starting from 0 every time.

PROBLEM 21 (*) Prove that every positive number can be written as the sum of distinct powers of 2 (including $2^0 = 1$).

PROBLEM 22 (*) Ten lamps are arranged in a line, all turned off. You may pick a lamp to toggle between on and off, but it will also toggle its immediate neighbors. Is it possible to only have the 5th lamp from the left on? Prove or disprove.

PROBLEM 23 (*) Prove that $n^5 - n$ is divisible by 30 for every positive integer n . Hint: You may use Fermat's little theorem: If p is prime, for any integer a we have $p|a^p - a$.

PROBLEM 24 (*) Prove that, among any $n + 1$ integers, not necessarily consecutive, at least two have a difference divisible by n .

PROBLEM 25 (*) Show that $1 + 3 + 9 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$ without induction.

Challenging Problems

PROBLEM 26 (**) Prove that any integer sequence of length n contains a subsequence (i.e. a consecutive subset) with sum divisible by n .

PROBLEM 27 (**) Prove that the decimal expansion of a rational number will eventually repeat.

PROBLEM 28 (**) Prove that there is no smallest positive rational number.

PROBLEM 29 (**) Prove that for all real numbers $x_1, \dots, x_n$, and for all positive integers n , then

$$(x_1 + \dots + x_n)^2 \leq n(x_1^2 + \dots + x_n^2).$$

PROBLEM 30 (**) Let $\alpha \in \mathbb{R}$ such that $\alpha + \frac{1}{\alpha} \in \mathbb{Z}$. Prove that $\alpha^n + \frac{1}{\alpha^n} \in \mathbb{Z}$ for any $n \in \mathbb{N} \cup \{0\}$.

Some AM-GM Identities

Please do the problems in this section in order.

PROBLEM 31 (**) This is known as the inequality of arithmetic and geometric means (AM-GM). For all positive real numbers a and b ,

$$\frac{a+b}{2} \geq \sqrt{ab}.$$

Hint: $(\sqrt{a} - \sqrt{b})^2$.

PROBLEM 32 Prove that for real a, b ,

$$a^4 + b^4 \geq 2a^2b^2.$$

Hint: AM-GM

PROBLEM 33 Prove that if $a, b, c > 0$ are real numbers, then

$$(a + b)(b + c)(c + a) \geq 8abc.$$

Hint: AM-GM.

PROBLEM 34 (*) Prove that if $a, b, c > 0$ are real numbers, then

$$\frac{a}{b} + \frac{b}{c} + \frac{c}{a} \geq 3.$$

Hint: AM-GM.

PROBLEM 35 (*) Prove that if $a, b, c > 0$ are real numbers, then

$$(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9.$$

Hint: AM-GM

PROBLEM 36 (*) Prove that if $a, b, c > 0$ are real numbers, then

$$\frac{a}{a+b} + \frac{b}{b+c} + \frac{c}{c+a} \geq 1.$$

Hint: Do not use AM-GM.

PROBLEM 37 (**) Given $a + b = 1$, prove

$$ab \leq \frac{1}{4} \text{ and } a^2 + b^2 \geq \frac{1}{2}.$$

Hint: Use AM-GM on a^2 and b^2 .

PROBLEM 38 (**) Given positive real numbers a, b, c with $abc = 1$, prove

$$a + b + c \geq 3.$$

Hint / Challenge: Use only 2-variable AM-GM.

PROBLEM 39 (****) Prove that for positive real numbers $x_1, \dots, x_n$,

$$\frac{x_1 + \cdots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \cdots x_n}.$$

Hint: Do Problems 4 and 21 first. Can you combine the two proofs?

PROBLEM 40 (**) Prove that for positive real a, b ,

$$\frac{a^3}{b^2} + \frac{b^3}{a^2} \geq a + b.$$

Hint: Apply AM-GM on three numbers.